Lab Exercise 20- Deploy a Docker Application using AWS ECR and ECS

1. Objective

In this lab, you will learn how to:

- Build a container image using **Docker**
- Store the container image in **Amazon Elastic Container Registry**
- Deploy the container on **Amazon Elastic Container Service**
- Access the running application through a public endpoint

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2. Prerequisites

Before starting the lab, ensure you have:

- AWS account with appropriate permissions
- Docker installed
- AWS Command Line Interface installed and configured
- Basic understanding of containers
- Internet access

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3. Lab Scenario

A development team has built a small web application using containers. Your task as a DevOps engineer is to:

1. Containerize the application using **Docker**
2. Store the container image in **Amazon ECR**
3. Deploy the container to **Amazon ECS**
4. Verify the application is running successfully

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4. Lab Tasks

Task 1: Create a Docker Application

Create a simple application.

Example

index.html

```
Hello from AWS ECS Lab
```

Create a **Dockerfile**

```
FROM nginx

COPY index.html /usr/share/nginx/html/index.html
```

Task 2: Build the Docker Image

Build the image locally using **Docker**.

```
Docker build -t ecs-lab-app .
```

Verify the image.

```
docker images
```

Task 3: Create an ECR Repository

Create a repository in **Amazon ECR**.

Steps:

1. Open AWS Console
2. Go to **Amazon ECR**
3. Click **Create Repository**
4. Repository name:

```
ecs-lab-repository
```

Task 4: Authenticate Docker to ECR

Login to the ECR registry using **AWS CLI**.

Example:

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin <account-id>.dkr.ecr.us-east-1.amazonaws.com
```

Task 5: Tag the Docker Image

Tag the image to match the ECR repository.

Example:

```
docker tag ecs-lab-app:latest <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

Task 6: Push the Image to ECR

Push the image to the repository.

```
docker push <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

Verify the image in the **ECR console**.

Task 7: Create ECS Cluster

Navigate to **Amazon ECS**.

Steps:

1. Click **Create Cluster**
2. Select **Fargate**
3. Provide cluster name:

```
ecs-lab-cluster
```

Task 8: Create Task Definition

Create a task definition.

Settings:

Container image:

```
<account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

```
Port mapping:

80
```

Task 9: Run ECS Service

Deploy the container.

Steps:

1. Create service
2. Select **Fargate**
3. Desired tasks:
4. Enable public IP

Task 10: Verify Deployment

After deployment:

1. Open ECS service
2. Copy the **public IP or load balancer URL**
3. Open in browser

Expected output:

```
Hello from AWS ECS Lab
```

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5. Expected Outcome

After completing this lab:

- Docker image is stored in **Amazon ECR**
- Container is running on **Amazon ECS**
- Application is accessible through a browser

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>mkdir ecs-lab-app

C:\Users\NK>cd ecs-lab-app

C:\Users\NK\ecs-lab-app>echo Hello from AWS ECS Lab > index.html

C:\Users\NK\ecs-lab-app>(echo FROM nginx & echo COPY index.html /usr/share/nginx/html/index.html) > Dockerfile

C:\Users\NK\ecs-lab-app>docker build -t ecs-lab-app .
[+] Building 1.1s (7/7) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 100B                                                                                0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                                     0.1s
 => [internal] load .dockerignore                                                                                   0.1s
 => => transferring context: 2B                                                                                     0.0s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 62B                                                                                    0.0s
 => CACHED [1/2] FROM docker.io/library/nginx:latest@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f  0.1s
 => => resolve docker.io/library/nginx:latest@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf  0.1s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                          0.1s
 => exporting to image                                                                                              0.5s
 => => exporting layers                                                                                             0.2s
 => => exporting manifest sha256:90637c9e527c9ebc3d198ee7b0fbfc3622357c0e97b0462ab37719c53e801123                  0.0s
 => => exporting config sha256:0c1afa49845a8533c69c4066b8f0930c6c56b1a12f51b4268442c81a32ee3d9a                    0.0s
 => => exporting attestation manifest sha256:832db7a0287fccc8406ad6370a26aed71a6f31a86a9e8beaa020b622b5b340be      0.1s
 => => exporting manifest list sha256:a1bf71617ae7a544d32d7c412854d4f9af7850040234afa1d02a1c2f0eac89b0             0.0s
 => => naming to docker.io/library/ecs-lab-app:latest                                                               0.0s
 => => unpacking to docker.io/library/ecs-lab-app:latest                                                            0.1s
```

```
C:\Users\NK\ecs-lab-app>docker images
                                                                                    i Info →   U   In Use
IMAGE                                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
12345:latest                                                         077e1c560bb9       286MB         86.6MB    U
busybox:latest                                                       b3255e7dfbcd      6.77MB         2.22MB
cmd-example:latest                                                   b93e22d50827       232MB         73.9MB    U
combined-example:latest                                              33f6b944b8d3       232MB         73.9MB    U
ecs-lab-app:latest                                                   a1bf71617ae7       237MB         62.9MB
entrypoint-example:latest                                            0786e2a71950       232MB         73.9MB    U
gcr.io/k8s-minikube/kicbase:v0.0.50                                  b97074569ae9      1.94GB          545MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93
                                                                     eb4fec00e8ad      1.94GB          545MB    U
krishk97/12345:latest                                                077e1c560bb9       286MB         86.6MB    U
mysql:latest                                                         c9e48b0c008f       1.3GB          290MB    U
nginx-html-app:latest                                                2548991e245e       237MB         62.9MB
nginx:latest                                                         c881927c4077       240MB         65.7MB
run-example:latest                                                   d37251da54e3       232MB         73.9MB
wordpress:latest                                                     054e61133439      1.08GB          275MB    U

C:\Users\NK\ecs-lab-app>
```